



THE KING'S SCHOOL

2005
Higher School Certificate
Trial Examination

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided
- All necessary working should be shown in every question

Total marks – 84

- Attempt Questions 1-7
- All questions are of equal value

Total marks – 84

Attempt Questions 1-7

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Marks

Question 1 (12 marks) Use a SEPARATE writing booklet.

(a) Find $\int \frac{dx}{\sqrt{9-4x^2}}$ **2**

- (b) $P(3,7)$ divides the interval AB in the ratio $m:n$,
where $A = (-1,2)$ and $B = (11,17)$.

Find the ratio $m:n$. **2**

(c) Let $f(x) = \ln(\tan x)$, $0 < x < \frac{\pi}{2}$

Show that $f'(x) = 2\operatorname{cosec}2x$ **3**

- (d) Two lines have gradients $2 + \sqrt{3}$ and 1 .

Find the size of the acute angle between these lines. **2**

(e) Use the substitution $u = 1 + x^2$ to evaluate $\int_1^{\sqrt{3}} 6x\sqrt{1+x^2} \, dx$ **3**

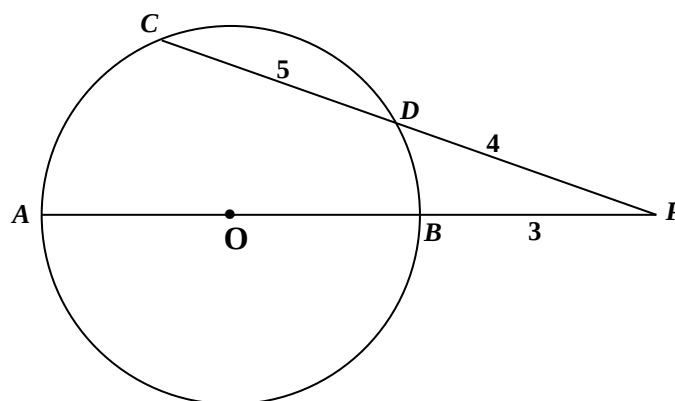
End of Question 1

(a) (i) Use the expansion of $\cos(A+B)$ to show that $\cos 2x = 1 - 2\sin^2 x$ 2

(ii) Evaluate $\lim_{x \rightarrow 0} \frac{\cos 2x - 1}{2x^2}$ 2

(b) Find $\int 4\sin^2 x \, dx$ 2

(c)



AB is the diameter of a circle, centre O .

AB produced meets the secant CD at P .

$CD = 5$, $DP = 4$ and $BP = 3$

Find the diameter of the circle. 2

(d) (i) Solve $\frac{x+1}{x^2+1} > 1$ 2

(ii) Without solving, explain why $x < -1$ is a solution of $\frac{x^2+1}{x+1} < 1$ 1

(iii) Hence, or otherwise, solve $\frac{x^2+1}{x+1} < 1$ 1

End of Question 2

- (a) The roots of $x^3 - 6x + 1 = 0$ are α , β , γ .

Find values for

(i) $\alpha + \beta + \gamma$ **1**

(ii) $\alpha^2 + \beta^2 + \gamma^2$ **2**

- (b) Use the substitution $t = \tan \frac{x}{2}$ to solve $\sin x - 7 \cos x - 5 = 0$, $0^\circ < x < 360^\circ$, giving your answers to the nearest degree. **3**

- (c) (i) A particle is moving along the x axis so that at any time $t \geq 0$ its velocity is v and its acceleration is $\frac{dv}{dt}$.

Prove that $\frac{dv}{dt} = \frac{d\left(\frac{1}{2}v^2\right)}{dx}$ **2**

- (ii) A particle is moving along the x axis in simple harmonic motion with its velocity v given by the equation $v^2 = 4x - 2x^2$.

Find

(α) its amplitude **2**

and (β) its period. **2**

End of Question 3

(a) $f(x) = 5\sqrt{1+x^6} - 7x$ has a zero near $x = 1$.

Use Newton's method once to find a two decimal place approximation to this zero.

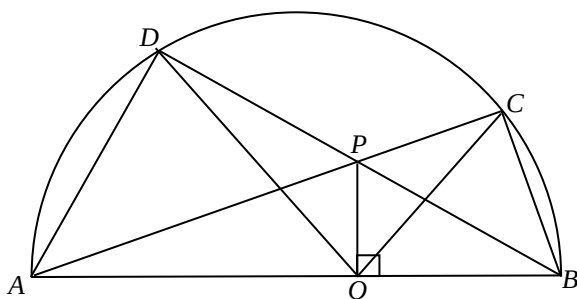
3

(b) Use mathematical induction to prove that for all integers $n \geq 1$,

$$(1^2 + 1)1! + (2^2 + 1)2! + \dots + (n^2 + 1)n! = n(n+1)!$$

4

(c)



AB is the diameter of a semi-circle $ABCD$.

DB meets AC at P and $PQ \perp AB$ at Q .

(i) Explain why $\angle DAC = \angle CBD$

1

(ii) Prove that $QBCP$ is a cyclic quadrilateral.

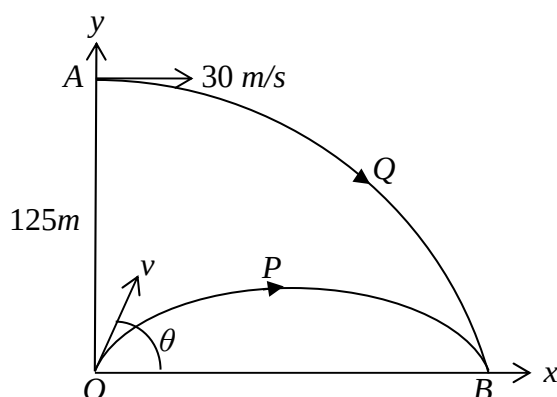
2

(iii) Deduce that PQ bisects $\angle DQC$.

2

End of Question 4

(a)



Two particles P and Q are projected at the same time in the same vertical plane. The only force acting on the particles is due to gravity, where $g = 10 \text{ m/s}^2$.

Taking axes of reference as in the diagram, particle P is projected from O on horizontal ground with velocity $v \text{ m/s}$ at an elevation of θ and particle Q is projected horizontally at 30 m/s from a point A 125 m above O . The two particles meet at the same time t at a point B on ground level.

You may assume the equations of motion for particle P are:

$$\begin{aligned}\ddot{x} &= 0 \\ \dot{x} &= v \cos \theta \\ x &= (v \cos \theta) t\end{aligned}$$

$$\begin{aligned}\ddot{y} &= -10 \\ \dot{y} &= -10t + v \sin \theta \\ y &= -5t^2 + (v \sin \theta) t\end{aligned}$$

[DO NOT PROVE THESE]

- | | |
|--|---|
| (i) Write down the equations of motion for particle Q | 2 |
| (ii) Find the time taken for particle Q to reach B and determine the horizontal distance it has travelled. | 2 |
| (iii) Find the velocity of projection and angle of elevation for particle P , i.e. find v and θ | 3 |
| (iv) Find the cartesian equation of the parabolic path of particle P . | 2 |

Question 5 continues on the next page

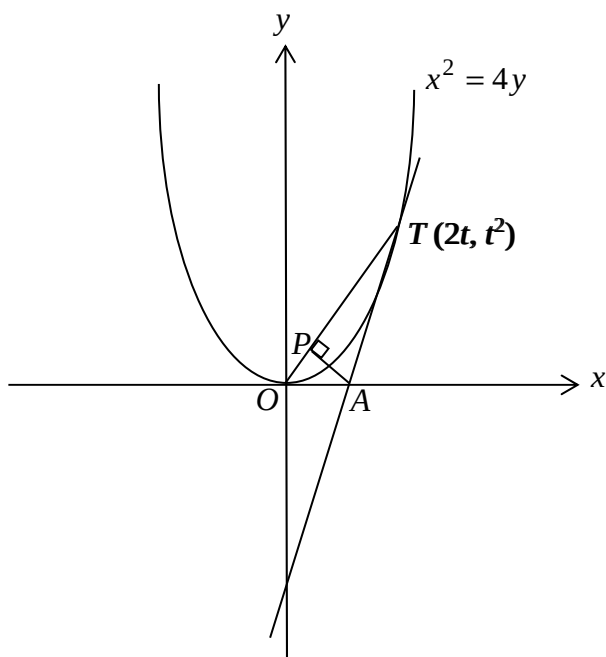
(b) You are given $P(x) = (2x^2 - x - 3)^3 \equiv 8x^6 - 12x^5 - 30x^4 + 35x^3 + Ax^2 - 27x - 27$

(i) Find the value of A . 1

(ii) Find two of the roots of $P''(x) = 0$ 2

End of Question 5

(a)



The tangent at $T(2t, t^2)$, $t \neq 0$, on the parabola $x^2 = 4y$ meets the x axis at A .

$P(x, y)$ is the foot of the perpendicular from A to OT , where O is the origin.

The equation of the tangent at T is $y = tx - t^2$

(i) Prove that the equation of AP is $y = -\frac{2}{t}(x - t)$ 2

(ii) Show that the equation of OT is $t = \frac{2y}{x}$ 1

(iii) Hence, or otherwise, prove that the locus of $P(x, y)$ lies on a circle with centre $(0,1)$ and give its radius. 3

Question 6 continues on the next page

- (b) The surface area S of a spherical bubble is changing at a constant rate of $k \text{ cm}^2 / \text{s}$.

Prove that the volume is changing at a rate proportional to the radius at any time t .

$$\left[S = 4\pi r^2, \quad V = \frac{4}{3}\pi r^3 \right] \quad 3$$

- (c) If $\log_6 9 = A$, express $\log_3 2$ in terms of A . 3

End of Question 6

- (a) Consider the function $f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$
- (i) Show that $f(-x) = -f(x)$ 1
- (ii) Show that $f(x) = 1 - \frac{2}{e^{2x} + 1}$ 1
- (iii) Explain why $f(x) < 1$ for all values of x 1
- (iv) Hence, or otherwise, sketch $y = f(x)$ in the number plane. 2
- (v) Explain why the inverse function $f^{-1}(x)$ exists without any restriction on the domain of $f(x)$ 1
- (vi) Sketch $y = f^{-1}(x)$ in the number plane. 1
- (vii) For $y = f^{-1}(x)$, express y in terms of x 2
- (b) (i) Show that $\frac{\binom{3n}{k}}{\binom{3n}{k-1}} = \frac{3n-k+1}{k}$ 1
- (ii) Find the greatest term in the expansion of $\left(1 + \frac{x}{2}\right)^{3n}$, n a positive integer, when $x = 1$ 2

End of Examination

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

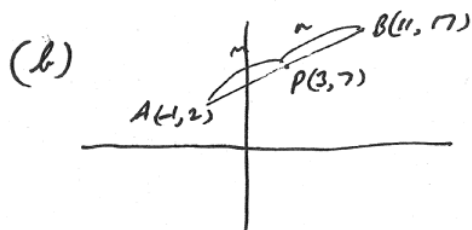
$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Ques 1 (a) $\frac{1}{2} \sin^{-1} \frac{2x}{3} \quad (+c)$



$\therefore m:n = 4:8 = 1:2$

(c) $f'(x) = \frac{\sec^2 x}{\tan x} = \frac{1}{\sin x \cos x}$
 $= \frac{2}{\sin 2x} = 2 \operatorname{cosec} 2x$

(d) $\tan \angle = \frac{2+\sqrt{3} - 1}{1 + 2+\sqrt{3}} = \frac{1+\sqrt{3}}{3+\sqrt{3}}$
 $= \frac{1+\sqrt{3}}{\sqrt{3}(1+\sqrt{3})} = \frac{1}{\sqrt{3}}$
 $\Rightarrow \angle = \frac{\pi}{6} \text{ or } 30^\circ$

(e) $u = 1+x^2$
 $\frac{du}{dx} = 2x$
 $\therefore x=1, u=2$
 $x=\sqrt{3}, u=4$

$\therefore I = \int_2^4 3\sqrt{u} \, du$
 $= 3.2 \left[\frac{u^{3/2}}{3/2} \right]_2^4$
 $= 2(8 - 2\sqrt{2})$
 $= 4(4 - \sqrt{2})$

Ques 2

$$(a) (i) \cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\Rightarrow \cos 2x = \cos^2 x - \sin^2 x \\ = 1 - \sin^2 x - \sin^2 x = 1 - 2\sin^2 x$$

$$(ii) \lim_{x \rightarrow 0} \frac{-2 \sin^2 x}{2x^2} = - \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^2 = -1$$

$$(b) I = 2 \int 1 - \cos 2x \, dx = 2 \left(x - \frac{\sin 2x}{2} \right) (+ c) \\ = 2x - \sin 2x \quad (+ c)$$

$$(c) \Rightarrow 3(d+3) = 9 \times 4 \\ \therefore d+3 = 12 \\ \text{ie diameter} = 9$$

$$(d) (i) \therefore x+1 > x^2+1 \\ \text{ie } x^2 - x < 0 \\ x(x-1) < 0 \\ \Rightarrow 0 < x < 1$$

$$(ii) x < -1 \Rightarrow x+1 < 0 \text{ and since } x^2+1 > 0, \\ \therefore \frac{x^2+1}{x+1} < 0 \Rightarrow \frac{x^2+1}{x+1} < 1$$

$$(iii) \text{ From (i) \& (ii), } x < -1 \text{ or } 0 < x < 1$$

Qn 3

(a) (i) $\alpha + \beta + \gamma = 0$

(ii) $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)$
 $= 0 - 2(-6) = 12$

(b) $t = \tan \frac{x}{2} \Rightarrow \frac{2t}{1+t^2} - \frac{7(1-t^2)}{1+t^2} - 5 = 0$

i.e. $2t - 7(1-t^2) - 5(1+t^2) = 0$

$\therefore 2t^2 + 2t - 12 = 0$

i.e. $t^2 + t - 6 = 0$

$(t-2)(t+3) = 0$

$\Rightarrow \tan \frac{x}{2} = 2 \text{ or } -3, 0^\circ < \frac{x}{2} < 180^\circ$

$\therefore x = 127^\circ \text{ or } 217^\circ$

(c) (i) $\frac{dv}{dt} = \frac{dv}{dx} \cdot \frac{dx}{dt}$

$= v \frac{dv}{dx}$

$= \frac{d(\frac{1}{2}v^2)}{dx} \cdot \frac{dv}{dx} = \frac{d(\frac{1}{2}v^2)}{dx}$

(ii) (a) $v = 0 \Rightarrow 2x(2-x) = 0$

i.e. $x = 0, 2$

$\therefore \text{amplitude} = 1$

(b) $\frac{1}{2}v^2 = 2x - x^2$

$\therefore \ddot{x} = 2 - 2x = -2(x-1)$

$\Rightarrow n^2 = 2 \text{ in usual notation}$

i.e. $n = \sqrt{2}$

$\therefore \text{period } T = \frac{2\pi}{\sqrt{2}} = \sqrt{2}\pi$

Ques 4

$$(a) f'(x) = 5 \cdot \frac{1}{2} (1+x^6)^{-\frac{1}{2}} \cdot 6x^5 - 7 = \frac{15x^5}{\sqrt{1+x^6}} - 7$$

$$\therefore x_1 = 1 - \frac{5\sqrt{2} - 7}{\frac{15}{\sqrt{2}} - 7} \approx 0.98$$

$$(b) \text{ For } n=1, \quad LS = (1^2+1)1! = 2$$
$$RS = 1(2!) = 2$$

$\therefore$ Assume $(1^2+1)1! + \dots + (n^2+1)n! = n(n+1)!$ for some integer $n \geq 1$

$$\text{Then } (1^2+1)1! + \dots + (n^2+1)n! + ((n+1)^2+1)(n+1)!$$

$$= n(n+1)! + (n^2+2n+2)(n+1)!, \text{ using the assumption}$$

$$= (n+1)! [n + n^2 + 2n + 2]$$

$$= (n+1)! (n^2 + 3n + 2)$$

$$= (n+1)! (n+1)(n+2)$$

$$= (n+1)(n+2)! = RS \text{ for } n+1$$

$\therefore \dots$

(c) (i) angles in the same segment

(ii) $\angle ACB = 90^\circ$, $\angle$ in a semi-circle

$$= \angle PQB, \text{ given}$$

$\therefore$ $QBPC$ is a cyclic $\square$, opp. angles supplementary

(iii) Similarly to (ii), $AQPD$ is a cyclic quad

$\therefore \angle PQD = \angle DAP$, $\angle$ s in same segment

$$+ \angle PQC = \angle CBP, \quad "$$

But $\angle DAP = \angle CBP$, given

$$\therefore \angle PQD = \angle PQC \text{ i.e. } PQ \text{ bisects } \angle DQC$$

Qn 5

$$\begin{array}{ll} \text{(a) (i)} & \ddot{x} = 0 \\ & \ddot{y} = -10 \\ & \dot{x} = 30 \\ & \dot{y} = -10t \\ & x = 30t \\ & y = -5t^2 + 125 \end{array}$$

$$\begin{aligned} \text{(ii) at B, } y=0 & \quad \therefore 5t^2 = 125 \\ & \quad \text{or } t^2 = 25 \quad \therefore t = 5 \\ & \quad \text{so } x = 30 \times 5 = 150 \\ & \quad \text{i.e. 5s to reach B 150m horizontally} \end{aligned}$$

$$\begin{aligned} \text{(iii) From (ii), } 150 &= 5v \cos \theta \quad \text{i.e. } v \cos \theta = 30 \\ \text{and } 0 &= -125 + 5v \sin \theta \quad \text{i.e. } v \sin \theta = 25 \\ \therefore v^2 &= 30^2 + 25^2 \Rightarrow v = \sqrt{30^2 + 25^2} \\ &= 5\sqrt{61} \text{ m/s} \quad \left[\begin{array}{l} \text{or } \sqrt{1525} \\ \text{or } 39.18 \dots \end{array} \right] \\ \text{and } \tan \theta &= \frac{25}{30} \Rightarrow \theta = \tan^{-1}\left(\frac{5}{6}\right) \quad \left[\begin{array}{l} \approx 39.8^\circ \\ \approx 0.7 \end{array} \right] \end{aligned}$$

$$\text{(iv) From (iii), } x = 30t, \quad y = -5t^2 + 25t$$

$$\begin{aligned} \therefore \text{Cartesian path is } y &= -5\left(\frac{x}{30}\right)^2 + 25\left(\frac{x}{30}\right) \quad [\text{will do}] \\ \text{i.e. } y &= -\frac{x^2}{180} + \frac{5x}{6} \end{aligned}$$

$$\begin{aligned} \text{(b) (i) Put } x &= -1, \text{ then } (2 + 1 - 3)^3 = 8 + 12 - 30 - 35 + A = 0 \\ \therefore A &= 45 \end{aligned}$$

$$\begin{aligned} \text{(ii) } P(x) &= (2x^2 - x - 3)^3 = ((x+1)(2x-3))^3 \\ \Rightarrow x &= -1 \text{ and } \frac{3}{2} \text{ are triple roots of } P(x) = 0 \\ \therefore \text{two roots of } P''(x) &= 0 \text{ are } x = -1, \frac{3}{2} \end{aligned}$$

Ques 6

(a) (i) Tangent at T is $y = tx - t^2$

$$\therefore \text{at } A, 0 = tx - t^2 \Rightarrow x = t$$

$$\text{i.e. } A = (t, 0)$$

$$\text{gradient of } OT = \frac{t^2}{2t} = \frac{t}{2}$$

$$\therefore AP \text{ is } y = -\frac{2}{t}(x - t)$$

$$(ii) \quad OT \text{ is } y = \frac{t}{2}x \quad \text{or } t = \frac{2y}{x}$$

(iii) From (ii), locus of $P(x, y)$ is

$$y = -\frac{2x}{2y} \left(x - \frac{2y}{x} \right)$$

$$\text{or } y^2 = -x^2 + 2y$$

$$\Rightarrow x^2 + y^2 - 2y = 0$$

$$\text{i.e. } x^2 + (y-1)^2 = 1,$$

a circle centre $(0, 1)$, radius 1 [except the point $(0, 0)$]

OTHERWISE using $y = -\frac{2}{t}(x - t)$ and $y = \frac{t}{2}x$

$$\text{we have } \frac{t}{2}x = -\frac{2}{t}(x - t)$$

$$\text{or } t^2x = -4(x - t) \Rightarrow x = \frac{4t}{t^2 + 4}$$

$$\therefore y = \frac{t}{2} \cdot \frac{4t}{t^2 + 4} = \frac{2t^2}{t^2 + 4}$$

$$\therefore x^2 + (y-1)^2 = \frac{16t^2}{(t^2 + 4)^2} + \left(\frac{2t^2}{t^2 + 4} - 1 \right)^2$$

$$= \frac{16t^2 + (t^2 - 4)^2}{(t^2 + 4)^2} = \frac{t^4 + 8t^2 + 16}{(t^2 + 4)^2} = \frac{(t^2 + 4)^2}{(t^2 + 4)^2} = 1$$

$\Rightarrow$ circle, centre $(0, 1)$, radius 1

$$(b) \quad \frac{dS}{dt} = k$$

$$\therefore \frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$$

$$= \frac{dV}{dr} \cdot \frac{dr}{dS} \cdot \frac{dS}{dt} \quad : \quad \frac{dS}{dr} = 8\pi r$$

$$= 4\pi r^2 \cdot \frac{1}{8\pi r} \cdot k$$

$$= \frac{k}{2} r$$

$\propto r$ since k is constant

$$(c) \quad \log_6 9 = \frac{\log_3 9}{\log_3 6} = \frac{2}{\log_3 3 + \log_3 2}$$

$$\Rightarrow A = \frac{2}{1 + \log_3 2}$$

$$\therefore 1 + \log_3 2 = \frac{2}{A}$$

$$\text{or } \log_3 2 = \frac{2}{A} - 1 \quad \text{or } \frac{2-A}{A}$$

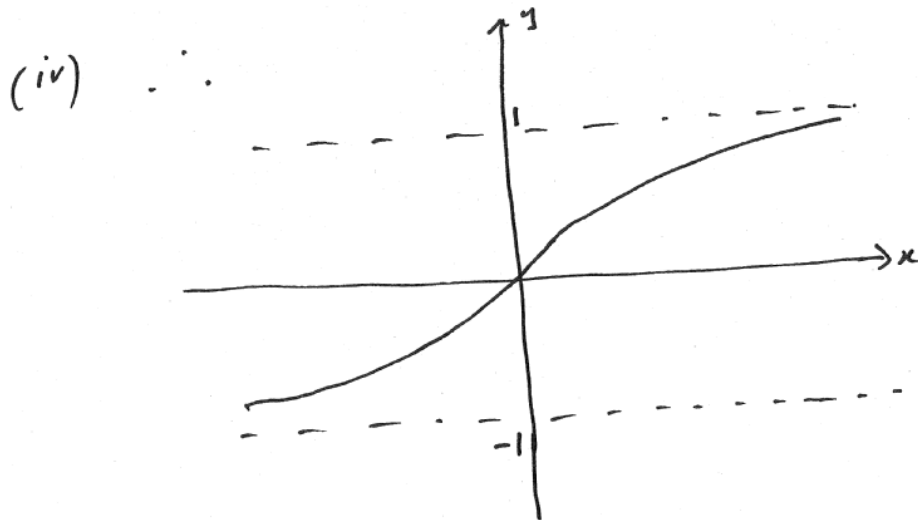
Qn 7

$$(a) (i) f(-x) = \frac{e^{-x} - e^x}{e^{-x} + e^x} = - \left(\frac{e^x - e^{-x}}{e^x + e^{-x}} \right) \\ = -f(x)$$

[$\Rightarrow$ odd function]

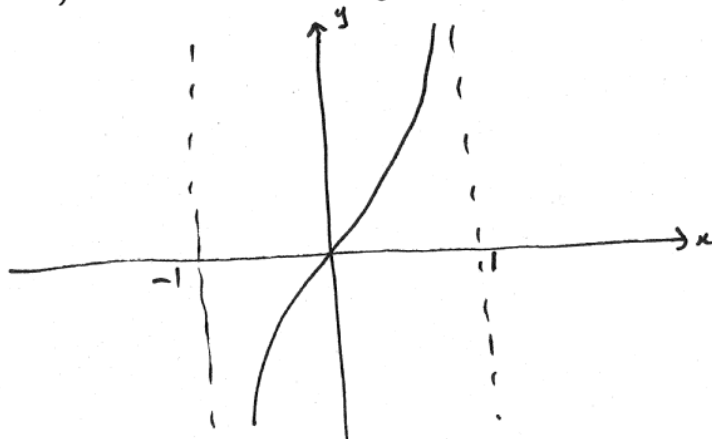
$$(ii) f(x) = \frac{e^{2x} - 1}{e^{2x} + 1} = \frac{(e^{2x} + 1) - 2}{e^{2x} + 1} \\ = 1 - \frac{2}{e^{2x} + 1}$$

$$(iii) \frac{2}{e^{2x} + 1} > 0 \quad \forall x \quad \therefore f(x) < 1 \text{ from (ii)}$$



(v) $\therefore f(x)$ is increasing $\forall x$

(vi)



$$(vii) f^{-1} : x = 1 - \frac{2}{e^{2y} + 1}$$

$$\Rightarrow \frac{2}{e^{2y} + 1} = 1 - x$$

$$\text{or } e^{2y} + 1 = \frac{2}{1-x}$$

$$\therefore e^{2y} = \frac{2}{1-x} - 1 = \frac{1+x}{1-x}$$

$$\therefore 2y = \ln\left(\frac{1+x}{1-x}\right)$$

$$\text{i.e. } y = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right)$$

$$(4) (i) \frac{\binom{3n}{k}}{\binom{3n}{k-1}} = \frac{(3n)! (3n-k+1)! (k-1)!}{(3n-k)! k! (3n)!}$$

$$= \frac{3n-k+1}{k}$$

$$(ii) \text{ Now, } \frac{u_{k+1}}{u_k} = \frac{3n-k+1}{k} \cdot \frac{x}{2} \text{ from (i)}$$

$$\therefore \text{ for } x=1, \frac{u_{k+1}}{u_k} > 1 \Rightarrow 3n-k+1 > 2k$$

$$\Rightarrow 3k < 3n+1$$

$$\text{or } k < n + \frac{1}{3}$$

$$\Rightarrow u_{k+1} > u_k \text{ for } k \leq n$$

$$\text{i.e. greatest term} = \binom{3n}{n} \frac{1}{2^n} \text{ when } x=1$$

$$[u_{n+1} \text{ will do}]$$